Research Brief — How Retrieval-Augmented Generation Works

Educational Test Document | Version 1.0

Document contents

Definition

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. Instead of relying only on a language model's stored parameters, a RAG system retrieves relevant source passages and provides them as context for generating an answer.

Typical pipeline

A common RAG pipeline has six stages: document ingestion, text extraction, chunking, embedding generation, vector indexing, and retrieval-augmented answer generation. Some systems add reranking, access-control filters, and citation verification.

Chunking

Chunking divides long documents into smaller passages. Very large chunks may contain irrelevant material, while very small chunks can lose useful context. Chunk size and overlap should be evaluated against representative questions rather than chosen by habit.

Embeddings and vector search

An embedding model converts text into numerical vectors. Similarity search compares the query vector with stored document vectors. FAISS is a library for efficient similarity search over dense vectors; it is not itself a language model.

Grounding and limitations

RAG can improve factual grounding when the retrieved passages contain the answer, but it does not eliminate hallucinations. A system should handle missing evidence explicitly, preserve source metadata, and evaluate retrieval quality separately from answer quality.

Evaluation

Useful retrieval metrics include Recall at k and Mean Reciprocal Rank. Answer-level evaluation can examine faithfulness, relevance, and citation correctness. A small, manually reviewed set of realistic questions is a practical starting point for local testing.

Test note: This is fictional sample content. It contains deliberately specific facts, dates, numbers, and names so retrieval and citation behavior can be checked.